

# Transparent Practices for Quantitative Empirical Research

Chat Wacharamanotham  
chat@acm.org  
Swansea University  
Swansea, United Kingdom

Xiaoying Pu  
xpu@ucmerced.edu  
University of California, Merced  
Merced, CA, USA

Fumeng Yang  
fy@northwestern.edu  
Northwestern University  
Evanston, IL, USA

Abhraneel Sarma  
abhraneel@u.northwestern.edu  
Northwestern University  
Evanston, IL, USA

## ABSTRACT

*Transparent research practices* enable the research design, materials, analytic methods, and data to be thoroughly evaluated and potentially reproduced. The HCI community has recognized research transparency as one quality aspect of paper submission and review since CHI 2021. This course presents current best practices and tools that increase research transparency for HCI researchers and students. The course will be **three online lectures** and **one in-person lab session**. The lectures will cover the most relevant concepts, guidelines, and practices in Open Science, frequentist statistics, Bayesian statistics, and uncertainty visualization. In the lab session, the course participants will interactively explore implications of analytical choices using RStudio in pairs or small group and receive tailored feedback from instructors. For any participants cannot come to Hamburg, we will provide a parallel online session.

## CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**; **Visualization techniques**; **Visualization design and evaluation methods**.

## KEYWORDS

transparent statistics, open science, uncertainty visualization, Bayesian statistics

### ACM Reference Format:

Chat Wacharamanotham, Fumeng Yang, Xiaoying Pu, and Abhraneel Sarma. 2023. Transparent Practices for Quantitative Empirical Research. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3574168>

## 1 MOTIVATION AND SCOPE

*“CHI papers should strive for research transparency regardless of the contribution type and methodology.”*

—CHI 2023 Guide to a successful submission [6]

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3574168>

*Transparent research practices* enable the research design, materials, analytic methods, and data to be thoroughly evaluated and potentially reproduced. The importance of transparent research practices is widely recognized. Research transparency is one of the guiding principles of the UNESCO Recommendation on Open Science—which is adopted by 193 member states around the world in 2021 [44]. As of the end of 2020, the Transparency and Openness Promotion (TOP) Guidelines [28] are implemented by 4,985 journals [26]—among them are prominent outlets from a variety of scholarly disciplines, e.g., *Science*, *Nature Human Behavior*, *Psychological Science*, *American Economic Association*.

In the last decade, the field of HCI has been continuously engaged in the conversation related to replicability and transparency (e.g., RepliCHI movement [52–55], Transparency movement [8, 19, 20, 46]). These efforts manifested in the community-led effort in 2019 [7] to add a section on research transparency to the *Guide to a successful submission and Reviewing*, which was officially adopted since CHI 2021 [4, 5]. However, other HCI outlets have yet to catch up: only 12% of HCI-related journals adopted the TOP guideline [1]. As for researchers' practices, a survey of CHI authors shows several misunderstandings about transparent practices for research materials and data [45]. The report by OECD points out that even seasoned researchers also need to acquire new Open Science skills [31, p. 92]. Education can improve the research practices of individuals (bottom-up) and expand community support for transparent research policies (top-down). However, the report by the U.S. National Academies of Sciences, Engineering, and Medicine indicates that such education is still rare in academic institutions [29, pp. 153–154]. This report also recommends that professional societies train students and researchers as well as supporting development of educational programs. This course was first offered at CHI 2022 [47] and was **the first training in transparent research practice or Open Science within SIGCHI and ACM**. In this second version of the course, we will present updated and streamlined materials and will take advantage of an in-person session—with an online backup—to facilitate learning.

The field of HCI has a wide variety of methods to acquire knowledge: quantitative, qualitative, design, and engineering—to name a few [33]. **The scope of this course focuses on quantitative empirical research.** A large body of research from the field of HCI and visualization had also contributed to improving transparent practices in quantitative research (e.g., [9, 12, 13, 15, 21, 22, 39, 42]). **This course will present current best practices and tools for research transparency.**

## 2 INTENDED AUDIENCE

The intended audience is HCI researchers and students who already have elementary knowledge in quantitative research design and statistical analysis. For example, they should already know the differences between the within-subjects and between-subjects design. They should have performed statistical analysis on a few datasets. They should also be familiar with how research articles report their method and results.

## 3 PREREQUISITES

In the course, we will use R to demonstrate analytic decisions, statistical analysis methods, and visualization techniques. The knowledge and experience of R will be beneficial for future adoption. The participants may have a better learning experience if they previously used an interpreted language (e.g., Python, R). Nevertheless, the participants are not expected to be proficient at R to understand the course materials. Several weeks before the course, we will send participants instructions for setting up R on their system and will provide a drop-in hour for technical support.

## 4 BENEFITS AND LEARNING OUTCOMES

After the course, we expect the participants to be able to get started on improving their own research practices, assessing research transparency in articles they read and review, as well as evaluating institutional policies that may impact research transparency. Toward these goals, we design this course with the following specific learning outcomes:

The course participants...

- are aware of decisions made in the course of quantitative research and the importance of making them transparent,
- will know a range of practices, methods, and tools to improve research transparency,
- can apply principles to evaluate and compare the transparency of statistical and visualization techniques, and
- can preregister and deposit their research artifacts in FAIR repositories.

## 5 CONTENT AND PRACTICAL WORK

Research transparency spans from planning to sharing results. We have organized it into four sections.

### 5.1 Planning research and sharing research artifacts

Planning involves deciding on the type of research questions, choosing variables to study, defining which variables and how they will be measured, and deciding on how many participants to recruit. Such decisions affect how to collect and analyze data. Ambiguities in describing the plan impede attempts to verify, replicate, and build upon the findings.

This section will introduce the participants to lists of research decisions (e.g., [49]). We will show examples of how these decisions could influence data analysis. There will be hands-on practice in identifying ambiguities and omissions of research decisions from excerpts of research papers. We will discuss strategies for planning

sample sizes [23], and the participants will explore how their decisions impact the statistical power using a simulation tool (e.g., [48]).

We will demonstrate how to preregister these research and analytical decisions. We will show several examples of preregistration and final research to illustrate the fact that “preregistration is a plan, not a prison” [11]—changes and further exploratory analyses are possible and can be made transparent. We will discuss types of research artifacts and how to share them according to the FAIR principles (Findability, Accessibility, Interoperability, and Reuse). We will also discuss ethical concerns in data collection and sharing.

### 5.2 Choosing statistical methods and reporting their results

This course recognizes that many statistical approaches are valid, and each approach has its own merits and challenges. Instead of presenting *one best method*, we will present a set of guiding principles [43] (provided in Appendix A) that allow the course participants to evaluate transparency. We will sample several statistical techniques and apply them to example datasets. Participants will have access to pre-written analysis R scripts. They will be able to experiment with different analytic decisions and discuss the implications from the perspective of transparency with their classmates.

For **frequentist statistics** (e.g., *t*-test, Wilcoxon, or ANOVA), we will discuss common mistakes in interpreting p-values and confidence intervals. We will discuss the differences in the understandability and usability between simple and standardized effect sizes. We will also discuss the implications of the presence and absence of pre-study power analyses. We will cover important statistical results that are necessary for meta-analysis.

For **Bayesian statistics**, we will start by introducing Bayesian statistical concepts, including prior and posterior distributions as well as the likelihood function. We will then walk through the implementation of several Bayesian models, and compare them with the frequentist analogs, followed up by extensions to these models with respect to increasing transparency. If time permits, we will also discuss other opportunities: (1) choosing prior distributions to improve both the models and transparency in the results; (2) using metrics such as information criteria to disclose transparency in model evaluation and selection; and (3) following best practices such as posterior predictive checks to increase transparency in reviewing processes.

### 5.3 Visualizing the results

Continuing from our session on statistical methods, we will equip participants with mindsets and tools to create transparency-oriented visualizations during their analysis as well as for their papers. Such visualizations for analysis results can make a paper more transparent in multiple ways, including improving faithfulness, robustness, process transparency, and clarity (which are transparent statistics principles, see Appendix A).

We will first introduce the idea that visualizing uncertainty information is staying faithful to their statistical analysis. Without proper uncertainty representation, we risk exaggerating the certainty of our findings [16]. Drawing from recent literature [34, 35], we will show different uncertainty visualization designs and the

concepts behind the variations. Notably, there are no one-size-fits-all solutions for uncertainty communication. Thus, we will teach the practice of selecting the most ethical and transparent option at each decision point within the data visualization pipeline. These concepts include frequency vs. probability framing, implicit vs. explicit uncertainty, and aleatory vs. epistemic uncertainty. Then, we will include an interactive activity where participants interpret frequency-based plots and discuss the potential pros and cons of example visualizations based on the concepts they just learned. We will show how multiverse analyses can increase the robustness of statistical analysis [13], and techniques for visualizing their results.

## 5.4 Practicing transparent research

The goal of the last session is to let participants practice what they learned in a lab activity. We will offer a detailed lab handout with guidelines, examples, datasets, and coding stencils prepared, and participants will explore, code, and discuss with us during this session.

The first half of the session aims to familiarize the participant with the processes involved in implementing and extracting results, such as summary statistics or posterior probability distributions, from a statistical model. We will provide participants with scaffolded code for implementing statistical analysis using R, lme4 [2], and brms [3]—from the first and the second session.

The second half of the session focuses on how to visualize statistical results in transparent manner. The content includes techniques that they have seen in the second and the third session: summary plots, plots for confidence distributions (frequentist) or posterior distributions (Bayesian), and other model diagnostic plots. We will provide partially completed visualization code using ggplot2 [50] and ggdist [18] R packages. In addition, we will offer practical guidelines for making visualizations as clear and simple as possible, especially for publication in research manuscripts [17]. These guidelines include “letting the white space speak”, “establishing viewing order”, and “matching effective visual encoding with importance.” Participants will benefit the most if they are able to apply what they would have learned immediately in the presence of instructors when they can ask questions.

## 5.5 Course sessions

These three topics will be organized into the following four sessions:

- Session 1 covers the first topic and frequentist statistics. (Online)
- Session 2 introduces theories of Bayesian statistical analysis with demonstration. (Online)
- Session 3 introduces concepts and examples in visualization for transparency. (Online)
- Session 4 will be a hands-on lab session where participants pair up and immediately practice transparent research in a guided set of R code. (Hybrid)

The last session will be on-site with an online backup for participants who cannot or do not intend to be in Hamburg. At least one instructor will moderate the discussion and give feedback to each pair. We will draw collective lessons learned and discuss them in a plenary at the end of the course.

## 6 LIMITATIONS

We do not expect that after a four-session course, the participants will immediately transform their entire research practices to be transparent. Such transformation will require looking into specifics of research methods and application domains. This course will provide a **broad overview** on what to be cognizant of, **pointers** on where to learn in-depth, and **practical experience** of a range of important skills. This teaching strategy will provide a foundation that is adequately broad as well as samples of experiences that will pique participants’ curiosity to further their knowledge.

## 7 INSTRUCTOR BACKGROUND

**Chat Wacharamanotham** is a lecturer at Swansea University and a mandated instructor at the University of Zurich. The focus of his work is on understanding and developing tools for planning, reporting, reading, and sharing quantitative research [14, 30, 48, 56, 57]. He is also a co-organizer of the Transparent Statistics in Human-Computer Interaction group and the Dagstuhl Seminar on Transparent Quantitative Research as a User Interface Problem. He has seven years of experience teaching a research method course for graduate students. In 2019, he received the university-wide teaching award in recognition of “teaching, which stimulates dialogue between lecturers and students as well as exchange between students in the best possible way” [32].

**Fumeng Yang** is a postdoctoral fellow at Northwestern University. Her recent research focuses on uncertainty visualizations for the general public and user modeling through statistical and machine learning models. She served as a Student Volunteers chair for IEEE VIS and co-instructed the precedent series of the proposed course.

**Xiaoying Pu** is a postdoctoral researcher at the University of California, Merced. In her work, she takes a human-centered approach to help data workers communicate uncertainty and data analyses [36, 38]. She has organized a CHI 2021 SIG on visualization grammars [37] and studied the open science practice of preregistration [40]. Her website is [xiaoyingpu.github.io](https://xiaoyingpu.github.io).

**Abhraneel Sarma** is a Ph.D. candidate at Northwestern University. His research interests include studying how people make decisions using visualizations, and how visualizations can be used for improving statistical analysis or reporting statistical results. In addition, he has studied how users implement certain aspects of Bayesian models [42] and has developed tools for conducting multiverse analysis which is an approach for more transparent statistical research [41].

## 8 RESOURCES

- Chapter 1 of the Transparent Statistics Guidelines [43] provides the full description of the principles, their rationale, and application examples.
- For the course participants who wish to expand their knowledge on estimation-approach in frequentist statistics, we recommend Cumming & Calin-Jageman’s book *Introduction to the New Statistics* [10].
- For the course participants who wish to familiarize themselves with Bayesian statistics, we recommend skimming

Betancourt's article *Towards A Principled Bayesian Workflow* [27] This article provides instructive visualizations and R codes. And for those who wish to dig deeper, we recommend McElreath's book *Statistical rethinking* [25] or Lambert's book *A student's guide to Bayesian statistics* [24].

- For the course participants who wish to learn R and ggplot, we recommend the *R for Data Science* [51]—which is available online free of charge. R proficiency is not a prerequisite for this course.

## 9 ACCESSIBILITY

The content of our course necessitates many data visualizations. We regret that such content may impose a barrier for people with visual impairments. For people with hearing impairment, we requested an automatic real-time transcription (e.g., those provided in Zoom) in this course proposal. We are not aware of further accessibility barriers.

## ACKNOWLEDGMENTS

We thank Lace Padilla and Matthew Kay for allowing us to use parts of their work as teaching materials for this course.

## A TRANSPARENT STATISTICS GUIDING PRINCIPLES

Quoted and abbreviated from [43]).

- (1) **Faithfulness:** A transparent statistical report should strive to capture and convey the “truth” as accurately as possible, especially concerning the uncertainty within the data.
- (2) **Robustness:** In order to minimize the likelihood of inaccurate (unfaithful) results, data analysis and reporting strategies that are robust to departures from statistical assumptions—or that make few assumptions—should ideally be preferred.
- (3) **Resilience:** Data analysis and reporting strategies should be resilient to statistical noise, i.e., they should yield similar outcomes across hypothetical replications of the same study.
- (4) **Process Transparency:** Data analysis and reporting strategies need to be explained rather than implied. The decisions made during the analysis and report writing should be communicated as explicitly as possible.
- (5) **Clarity:** Study reports should be easy to process—even when they target experts. Ideally study reports should be accessible to most members of the HCI community, instead of being comprehensible by only a handful of specialists.
- (6) **Simplicity:** When choosing between two data analysis procedures, the simplest procedure should ideally be preferred even if it is slightly inferior in other respects.
- (7) **Non-contingency:** When possible and outside exploratory analyses, data analysis and reporting strategies should avoid decisions that are contingent on data, e.g., “if the data turns out like this, compute this, or report that”.
- (8) **Precision and economy:** Even if full transparency is achieved, a study report where nothing conclusive can be said would be a waste of readers' time, and may prompt them to seek inexistent patterns. Data quality, high statistical power, and high statistical precision are important goals to pursue.

- (9) **Material availability:** Sharing as much study material as possible is a core part of transparent statistics, as it greatly facilitates peer scrutiny and replication.

## REFERENCES

- [1] Nick Ballou, Vivek R. Warriar, and Sebastian Deterding. 2021. *Are You Open? A Content Analysis of Transparency and Openness Guidelines in HCI Journals*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445584>
- [2] Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. 2015. Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software* 67, 1 (2015), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- [3] Paul-Christian Bürkner. 2021. Bayesian Item Response Modeling in R with brms and Stan. *Journal of Statistical Software* 100, 5 (2021), 1–54. <https://doi.org/10.18637/jss.v100.i05>
- [4] CHI 2021 Organizers. 2021. *Guide to a Successful Submission*. <https://chi2021.acm.org/for-authors/presenting/papers/guide-to-a-successful-submission>
- [5] CHI 2021 Organizers. 2021. *Guide to Reviewing Papers*. <https://chi2021.acm.org/for-authors/presenting/papers/guide-to-reviewing-papers/>
- [6] CHI 2023 Organizers. 2022. *Guide to a Successful Submission*. <https://chi2023.acm.org/submission-guides/guide-to-a-successful-submission/#Transparency>
- [7] CHI guideline contributors in alphabetical order, Pernille Bjørn, Fanny Chevalier, Pierre Dragicevic, Shion Guha, Steve Haroz, Helen Ai He, Elaine M. Huang, Matthew Kay, Ulrik Lyngs, Joanna McGrenere, Christian Remy, Poorna Talkad Sukumar, and Chat Wacharamanotham. 2019. Proposal for changes to the CHI reviewing guidelines. <https://doi.org/10.5281/zenodo.5566172>
- [8] Lewis L. Chuang and Ulrike Pfeil. 2018. Transparency and Openness Promotion Guidelines for HCI. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI EA '18). ACM, New York, NY, USA, Article SIG04, 4 pages. <https://doi.org/10.1145/3170427.3185377>
- [9] Andy Cockburn, Carl Gutwin, and Alan Dix. 2018. HARK No More: On the Preregistration of CHI Experiments. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). ACM, New York, NY, USA, Article 141, 12 pages. <https://doi.org/10.1145/3173574.3173715>
- [10] Geoff Cumming and Robert Calin-Jageman. 2016. *Introduction to the new statistics: Estimation, open science, and beyond*. Routledge.
- [11] Alexander DeHaven. 2017. *Preregistration: A plan, not a prison*. <https://www.cos.io/blog/preregistration-plan-not-prison>
- [12] Pierre Dragicevic. 2016. *Fair Statistical Communication in HCI*. Springer International Publishing, Cham, 291–330. [https://doi.org/10.1007/978-3-319-26633-6\\_13](https://doi.org/10.1007/978-3-319-26633-6_13)
- [13] Pierre Dragicevic, Yvonne Jansen, Abhraneel Sarma, Matthew Kay, and Fanny Chevalier. 2019. Increasing the Transparency of Research Papers with Explorable Multiverse Analyses. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3290605.3300295>
- [14] Alexander Eiselmayer, Chat Wacharamanotham, Michel Beaudouin-Lafon, and Wendy E. Mackay. 2019. Touchstone2: An Interactive Environment for Exploring Trade-offs in HCI Experiment Design. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). ACM, New York, NY, USA, Article 217, 11 pages. <https://doi.org/10.1145/3290605.3300447>
- [15] Sebastian S. Feger, Sünje Dallmeier-Tiessen, Albrecht Schmidt, and Pawel W. Woźniak. 2019. Designing for Reproducibility: A Qualitative Study of Challenges and Opportunities in High Energy Physics. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3290605.3300685>
- [16] Jouni Helske, Satu Helske, Matthew Cooper, Anders Ynnerman, and Lonni Besancon. 2021. Can Visualization Alleviate Dichotomous Thinking Effects of Visual Representations on the Cliff Effect. *IEEE Transactions on Visualization and Computer Graphics* (2021), 1–1. <https://doi.org/10.1109/TVCG.2021.3073466> <https://ieeexplore.ieee.org/document/9405484/>
- [17] Matthew Kay. 2019. Visualization for scientific communication. (2019). DataViz Rackham Interdisciplinary Workshop.
- [18] Matthew Kay. 2021. *ggdist: Visualizations of distributions and uncertainty*. <https://doi.org/10.5281/zenodo.5115107>
- [19] Matthew Kay, Steve Haroz, Shion Guha, and Pierre Dragicevic. 2016. Special Interest Group on Transparent Statistics in HCI. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (CHI EA '16). ACM, New York, NY, USA, 1081–1084. <https://doi.org/10.1145/2851581.2886442>
- [20] Matthew Kay, Steve Haroz, Shion Guha, Pierre Dragicevic, and Chat Wacharamanotham. 2017. Moving Transparent Statistics Forward at CHI. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI EA '17). ACM, New York, NY, USA, 534–541. <https://doi.org/10.1145/3027063.3027084>

- [21] Matthew Kay, Tara Kola, Jessica R. Hullman, and Sean A. Munson. 2016. When (Is) is My Bus? User-Centered Visualizations of Uncertainty in Everyday, Mobile Predictive Systems. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 5092–5103. <https://doi.org/10.1145/2858036.2858558>
- [22] Matthew Kay, Gregory L. Nelson, and Eric B. Hekler. 2016. Researcher-Centered Design of Statistics: Why Bayesian Statistics Better Fit the Culture and Incentives of HCI. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 4521–4532. <https://doi.org/10.1145/2858036.2858465>
- [23] Daniel Lakens. 2021. Sample Size Justification (pre-print). <https://doi.org/10.31234/osf.io/9d3yf>
- [24] Ben Lambert. 2018. *A student's guide to Bayesian statistics*. Sage.
- [25] Richard McElreath. 2018. *Statistical rethinking: A Bayesian course with examples in R and Stan*. Chapman and Hall/CRC.
- [26] David Thomas Mellor, Jolene Esposito, Alexander C. DeHaven, and Victoria Stodden. 2019. TOP Resources - Evidence and Practices: TOP\_implementors.csv. <https://osf.io/kgvna/>
- [27] Michael Betancourt. 2020. *Towards A Principled Bayesian Workflow*. [https://betanalpha.github.io/assets/case\\_studies/principled\\_bayesian\\_workflow.html](https://betanalpha.github.io/assets/case_studies/principled_bayesian_workflow.html)
- [28] E Miguel, C Camerer, K Casey, J Cohen, K M Esterling, A Gerber, R Glennerster, D P Green, M Humphreys, G Imbens, D Laitin, T Madon, L Nelson, B A Nosek, M Petersen, R Sedlmayr, J P Simmons, U Simonsohn, and M Van der Laan. 2014. Promoting transparency in social science research. *Science (New York, N.Y.)* 343, 6166 (01 2014), 30–31. <https://doi.org/10.1126/science.1245317>
- [29] National Academies of Sciences, Engineering, and Medicine. 2018. *Open Science by Design: Realizing a Vision for 21st Century Research*. The National Academies Press, Washington, DC. <https://doi.org/10.17226/25116>
- [30] Natalia Obukhova. 2021. A Meta-Analysis of Effect Sizes of CHI Typing Experiments. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 476, 7 pages. <https://doi.org/10.1145/3411763.3451520>
- [31] OECD. 2015. Making Open Science a Reality. *OECD Science, Technology and Industry Policy Papers* 25 (October 2015). <https://doi.org/10.1787/5jrs2f963zsl-en>
- [32] University of Zurich. 2019. *Lehrpreis der Universität Zürich 2019 (Credit Suisse Award for Best Teaching)*. <https://www.uzh.ch/cmsssl/de/about/portrait/awards/teaching/2019.html>
- [33] Judith S Olson and Wendy A Kellogg (Eds.). 2014. *Ways of Knowing in HCI*. Springer. <https://doi.org/10.1007/978-1-4939-0378-8>
- [34] Lace Padilla, Racquel Fygenson, Spencer C Castro, and Enrico Bertini. 2022. Multiple Forecast Visualizations (MFVs): Trade-offs in Trust and Performance in Multiple COVID-19 Forecast Visualizations. *IEEE Transactions on Visualization and Computer Graphics* (2022).
- [35] Lace M. K. Padilla, Maia Powell, Matthew Kay, and Jessica Hullman. 2021. Uncertain About Uncertainty: How Qualitative Expressions of Forecaster Confidence Impact Decision-Making With Uncertainty Visualizations. *Frontiers in Psychology* 11 (2021), 3747. <https://doi.org/10.3389/fpsyg.2020.579267>
- [36] Xiaoying Pu and Matthew Kay. 2020. A Probabilistic Grammar of Graphics. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376466>
- [37] Xiaoying Pu, Matthew Kay, Steven M. Drucker, Jeffrey Heer, Dominik Moritz, and Arvind Satyanarayan. 2021. Special Interest Group on Visualization Grammars. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 162, 3 pages. <https://doi.org/10.1145/3411763.3450406>
- [38] Xiaoying Pu, Sean Kross, Jake M. Hofman, and Daniel G. Goldstein. 2021. *Data-mations: Animated Explanations of Data Analysis Pipelines*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445063>
- [39] Xiaoying Pu, Licheng Zhu, Matthew Kay, and Frederick Conrad. 2019. Designing for Preregistration: A User-Centered Perspective. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI EA '19). Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3290607.3312862>
- [40] Xiaoying Pu, Licheng Zhu, Matthew Kay, and Frederick Conrad. 2019. Designing for preregistration: A user-centered perspective. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [41] Abhraneel Sarma, Alexander Kale, Michael Moon, Nathan Taback, Fanny Chevalier, Jessica Hullman, and Matthew Kay. 2021. multiverse: Multiplexing Alternative Data Analyses in R Notebooks. *OSF Preprints* (2021). <https://osf.io/yfbwm>
- [42] Abhraneel Sarma and Matthew Kay. 2020. Prior Setting in Practice: Strategies and Rationales Used in Choosing Prior Distributions for Bayesian Analysis. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3313831.3376377>
- [43] Transparent Statistics in Human-Computer Interaction Working Group. 2019. *Transparent Statistics Guiding Principles*. <https://dx.doi.org/10.5281/zenodo.1186169>
- [44] UNESCO. 2021. UNESCO Recommendation on Open Science. <https://unesdoc.unesco.org/ark:/48223/pf0000379949.locale=en>
- [45] Chat Wacharamanatham, Lukas Eisenring, Steve Haroz, and Florian Ehtler. 2020. Transparency of CHI Research Artifacts: Results of a Self-Reported Survey. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376448>
- [46] Chat Wacharamanatham, Matthew Kay, Steve Haroz, Shion Guha, and Pierre Dragicevic. 2018. Special Interest Group on Transparent Statistics Guidelines. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI EA '18). ACM, New York, NY, USA, Article SIG08, 4 pages. <https://doi.org/10.1145/3170427.3185374>
- [47] Chat Wacharamanatham, Fumeng Yang, Xiaoying Pu, Abhraneel Sarma, and Lace Padilla. 2022. Transparent Practices for Quantitative Empirical Research. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 122, 5 pages. <https://doi.org/10.1145/3491101.3503760>
- [48] Xiaoyi Wang, Alexander Eiselmayer, Wendy E. Mackay, Kasper Hornbaek, and Chat Wacharamanatham. 2021. Argus: Interactive a priori Power Analysis. *IEEE Transactions on Visualization and Computer Graphics* 27, 2 (2021), 432–442. <https://doi.org/10.1109/TVCG.2020.3028894>
- [49] Jelte M. Wicherts, Coosje L. S. Veldkamp, Hilde E. M. Augusteijn, Marjan Bakker, Robbie C. M. van Aert, and Marcel A. L. M. van Assen. 2016. Degrees of Freedom in Planning, Running, Analyzing, and Reporting Psychological Studies: A Checklist to Avoid p-Hacking. *Frontiers in Psychology* 7 (2016), 1832. <https://doi.org/10.3389/fpsyg.2016.01832>
- [50] Hadley Wickham. 2016. *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. <https://ggplot2.tidyverse.org>
- [51] Hadley Wickham and Garrett Golemund. 2016. *R for data science: import, tidy, transform, visualize, and model data*. O'Reilly Media, Inc. <https://r4ds.had.co.nz/>
- [52] Max Wilson, Wendy Mackay, Ed Chi, Michael Bernstein, and Jeffrey Nichols. 2012. RepliCHI SIG: From a panel to a new submission venue for replication. In *CHI'12 Extended Abstracts on Human Factors in Computing Systems*. ACM, 1185–1188.
- [53] Max L Wilson, Ed H Chi, Stuart Reeves, and David Coyle. 2014. RepliCHI: the workshop II. In *CHI'14 Extended Abstracts on Human Factors in Computing Systems*. ACM, 33–36.
- [54] Max L Wilson, Wendy Mackay, Ed Chi, Michael Bernstein, Dan Russell, and Harold Thimbleby. 2011. RepliCHI-CHI should be replicating and validating results more: discuss. In *CHI'11 Extended Abstracts on Human Factors in Computing Systems*. ACM, 463–466.
- [55] Max LL Wilson, Paul Resnick, David Coyle, and Ed H Chi. 2013. RepliChi: the workshop. In *CHI'13 Extended Abstracts on Human Factors in Computing Systems*. ACM, 3159–3162.
- [56] Chia-Kai Yang, Tanja Blascheck, and Chat Wacharamanatham. 2020. A Comparison of a Transition-Based and a Sequence-Based Analysis of AOI Transition Sequences. In *ACM Symposium on Eye Tracking Research and Applications* (Stuttgart, Germany) (ETRA '20 Short Papers). Association for Computing Machinery, New York, NY, USA, Article 7, 5 pages. <https://doi.org/10.1145/3379156.3391349>
- [57] Chia-Kai Yang and Chat Wacharamanatham. 2020. Asymmetric Effect of Text-Chart Proximity on Reading Behavior. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society* (Tallinn, Estonia) (NordiCHI '20). Association for Computing Machinery, New York, NY, USA, Article 15, 9 pages. <https://doi.org/10.1145/3419249.3420184>